PART A

Experiment 1

A.1 Aim: Implement **Setting up an Agile Project Lifecycle in Jira: Epics, User Stories, and Sprint Planning.**

A.2 Prerequisite

- Basic knowledge of Software Development Life Cycle (SDLC)
- Fundamentals of Agile Methodology and Scrum Framework
- Understanding of Agile artifacts (Epic, User Story, Sprint, Backlog)
- Familiarity with project management concepts

A.3 Outcome

After successful completion of this experiment, students will be able to:

- Create and configure an Agile project in Jira.
- Create and manage Epics, User Stories, and Tasks.
- Organize and prioritize the Product Backlog.
- Plan and execute Sprint Planning activities.
- Track project progress using Jira Agile boards.

A.4 Theory

What is Setting up an Agile Project Lifecycle in Jira?

Setting up an Agile Project Lifecycle in Jira involves configuring a project to support Agile software development using Scrum or Kanban methodologies. Jira provides tools for creating and managing project artifacts such as Epics, User Stories, Tasks, Sprints, and Backlogs. It enables teams to plan, track, collaborate, and deliver software iteratively while maintaining transparency throughout the development lifecycle.

Explain the Application in Detail

Jira is one of the most widely used Agile project management platforms that helps software development teams organize work efficiently. It supports the complete Agile lifecycle from requirement gathering to project delivery.
Key applications include:

- Creating Scrum or Kanban projects
- Managing Product Backlogs
- Creating and organizing Epics
- Writing User Stories and Tasks
- Sprint Planning and Sprint Management
- Task assignment and workload distribution
- Progress tracking using Agile boards
- Burndown charts and Sprint Reports
- Collaboration among developers, testers, and project managers
- Issue tracking and workflow automation

Organizations use Jira to improve team collaboration, monitor project status, prioritize work items, and ensure timely software delivery.

Which Agile Framework, Tool, and Features are Used?

Framework

- Agile Software Development
- Scrum Methodology

Tool

- Atlassian Jira Software

Key Jira Features

- Agile Project Creation
- Scrum Board / Kanban Board
- Product Backlog
- Epics
- User Stories
- Tasks and Subtasks
- Sprint Creation
- Sprint Planning
- Sprint Backlog
- Workflow Management
- Dashboards and Reports

Common Agile Artifacts

- Epic
- User Story
- Task
- Bug
- Sprint
- Product Backlog
- Release

JQL (Jira Query Language) Examples

project = "AI Project"

issuetype = Story

status = "To Do"

assignee = currentUser()

Sprint in openSprints()

Common Workflow

To Do

↓

In Progress
↓
Code Review
↓
Testing
↓
Done

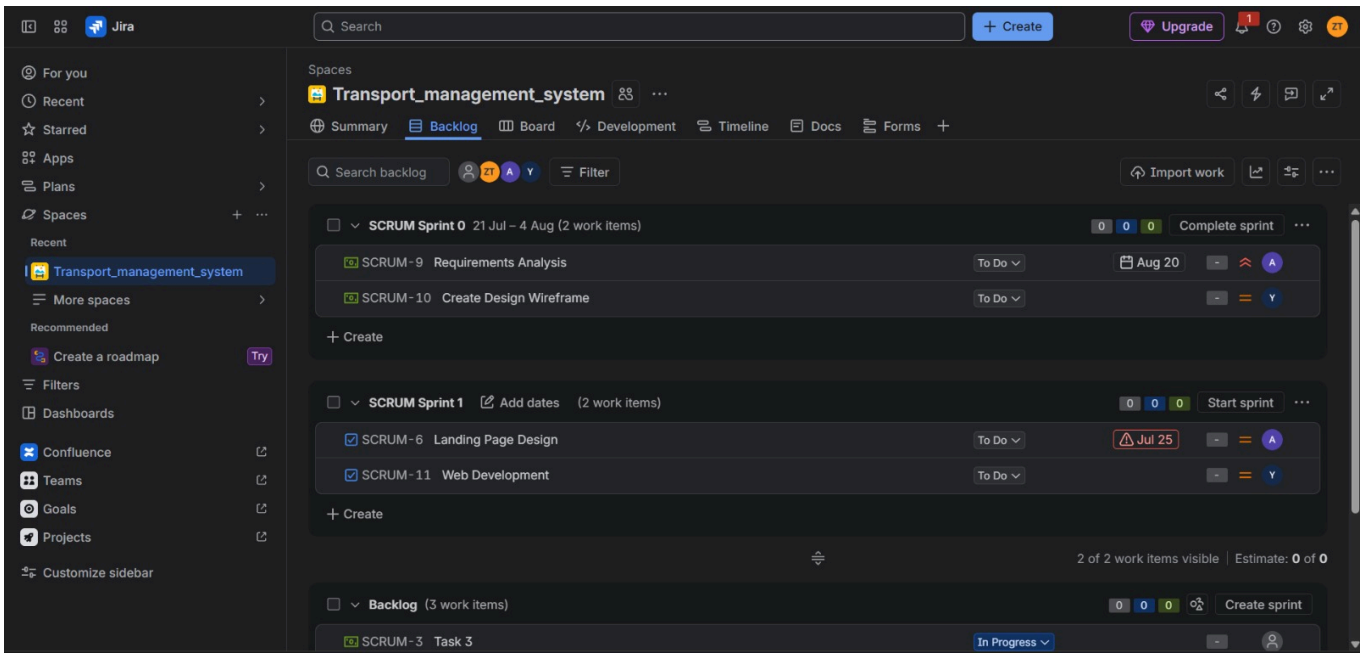
This experiment enables students to understand how Agile principles are implemented practically using Jira by creating Epics, decomposing them into User Stories, planning Sprints, and tracking project execution effectively.

PART B
(PART B: TO BE COMPLETED BY STUDENTS)

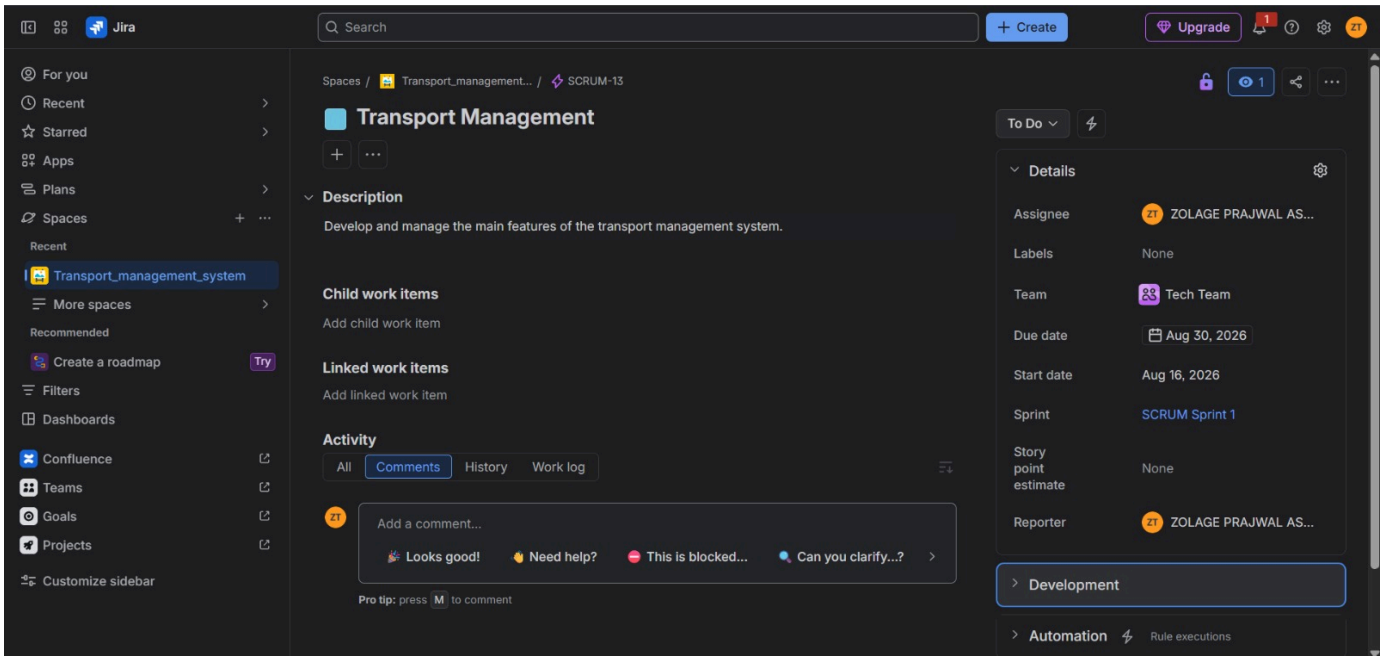
Roll No. A27	Name: Prajwal Ashok Zolage
Class: BE–AI&DS	Batch:A1
Date of Experiment:	Date of Submission:

B.1 Code Screenshot:

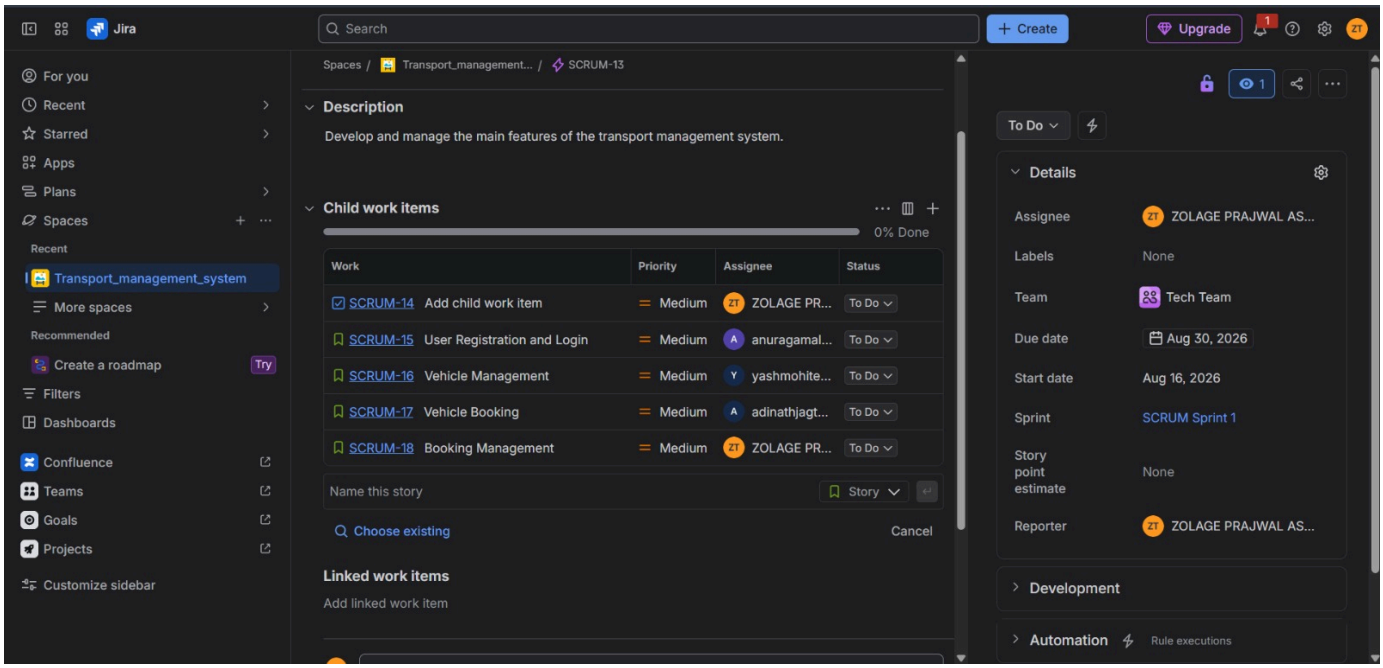
1. **Creation of Agile Scrum Project in Jira**



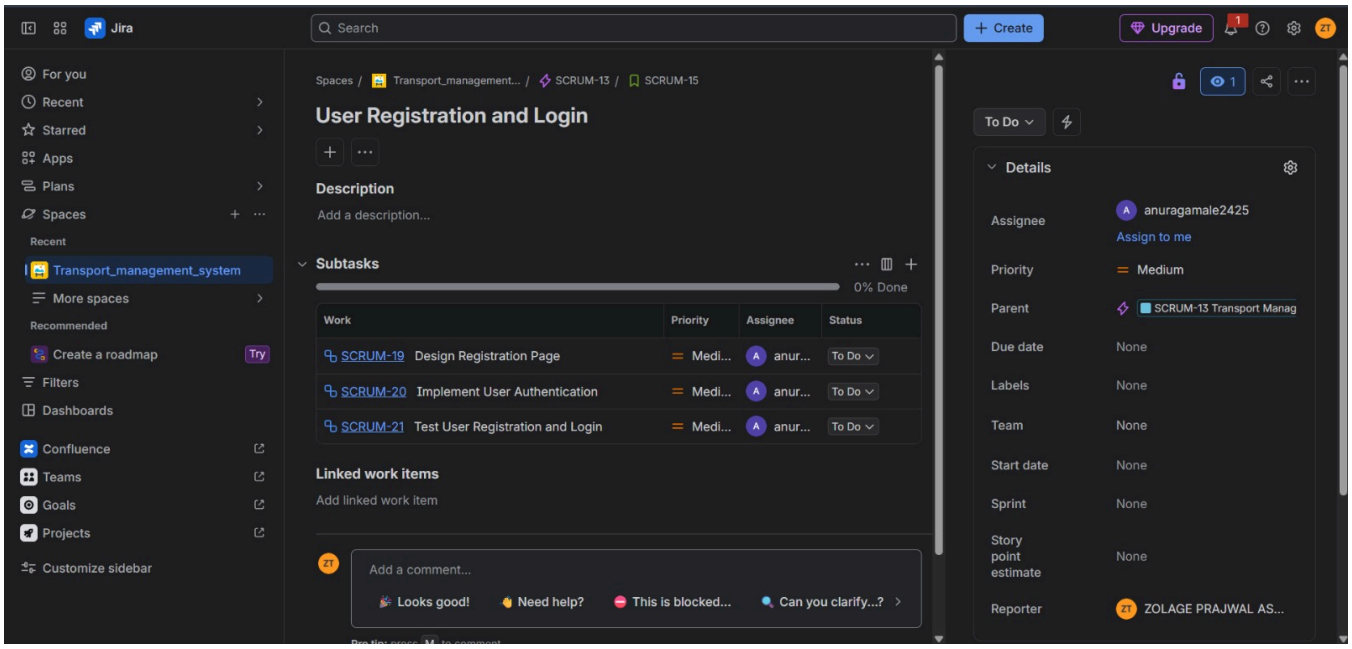
2. Creation of Epic in Jira



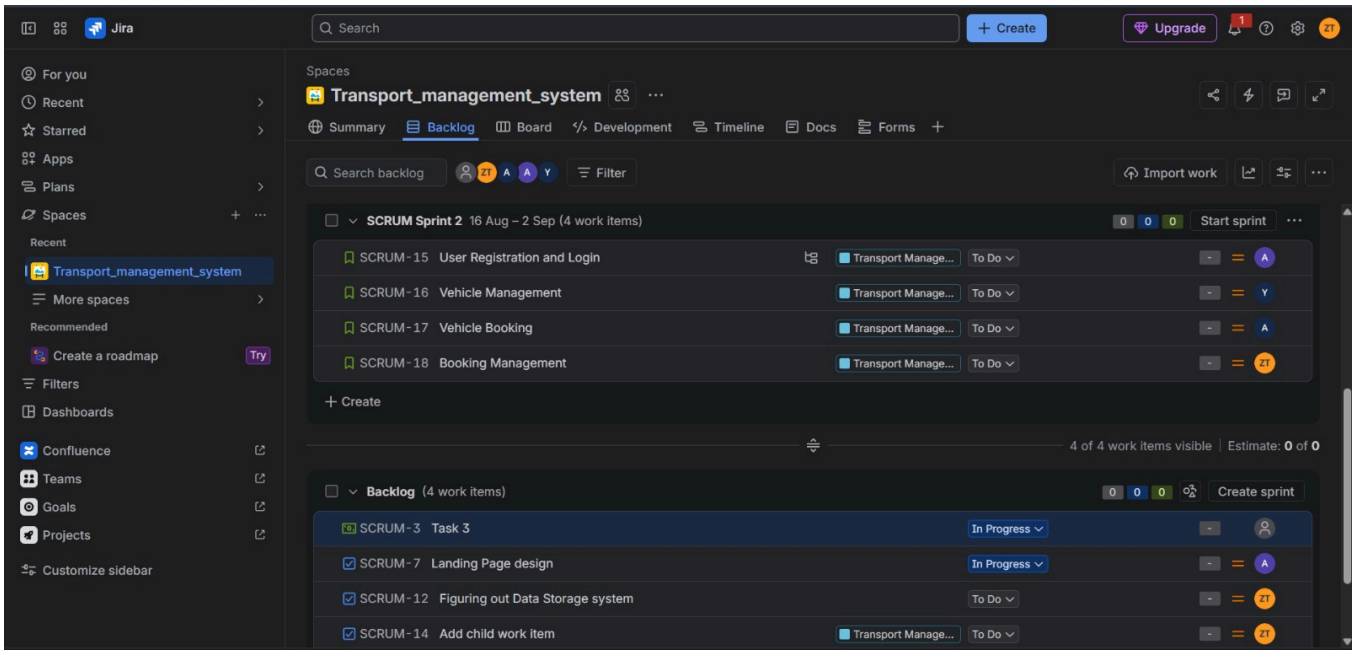
3. Creation of User Stories in Product Backlog



4. Creation of Tasks and Subtasks



5. Prioritized Product Backlog



B.2 Input and Output:
Input:

- Jira software project named **Transport Management System**.
- Scrum backlog containing user stories and tasks.
- Four user stories:
    1. User Registration and Login
    2. Vehicle Management
    3. Vehicle Booking
    4. Booking Management
- Assigned team members for the respective work items.
- Scrum Sprint 2 containing the selected user stories.
- Subtasks created under the user stories.

Output:

- A properly organized **Scrum Sprint and Product Backlog** in Jira.
- User stories assigned to team members.
- Subtasks associated with the respective user stories.
- Scrum Board showing the progress of work items through **To Do, In Progress, and Done** states.
- Successful demonstration of Agile/Scrum project management using Jira.

B.3 Conclusion:
The experiment was successfully completed using **Jira** to implement Agile Scrum practices for the Transport Management System. A Scrum backlog was created with user stories, team members were assigned to work items, and the stories were organized into a sprint. The Scrum board was used to manage and track the progress of tasks through different stages. Thus, the experiment demonstrated the practical use of **Agile methodology and Scrum project management using Jira**.